What holds a paper boat?

What holds a paper boat?

Name _____ Date _____



Little Wonderer set a sheet of paper beside a tub. “Could this carry a penny?” Big Wonderer put down a short list of folding commands. Little traced the first line. The paper was flat and flimsy. Somehow, those steps would have to give it a new shape.



“Fold the paper in half. Fold the corners down.” Little followed each command in order. More folds made an open boat with walls and a broad bottom. “Like a program,” Little said, “each step helps the next one work.” Now came the test. Would the folds hold?



Little lowered the boat into the tub. It floated: it stayed at the water’s surface. The boat pushed water aside, or displaced it. Water pushed back up. That push equaled the weight of the displaced water. When the upward push balanced the boat’s weight, the boat stopped sinking lower.

float	stay at the surface of water
displace	push water aside to make room
waterline	the line where the water meets the boat
load	the weight a boat carries
absorb	take water into a material
command	a sentence that tells what to do

What holds a paper boat?

What holds a paper boat?

Name _____ Date _____



Little added a penny, then another. This load was the weight the boat carried. The boat sank lower, displacing more water. Its waterline, where water met the paper, climbed up the sides. Folded walls kept water out. The broad bottom and air-filled space inside helped the boat displace enough water.



“Add one penny. Watch the waterline,” Little wrote. Then one more penny made the rim dip under. Water spilled inside. The boat was swamped! A flat sheet had no walls to protect an air-filled space. Shape mattered, and even this folded boat could not carry an endless load.



Later, the boat sagged. Paper’s tiny plant fibers absorb water: they take it in. Wet paper grows heavy and soft until its folds lose their shape. A crayon or wax coat slows soaking. Little saved the commands. “Could my next boat last longer?”

1. What pushed up on the boat, and what did the boat push aside?
2. Why did the waterline climb when Little added pennies? Use the word load.
3. How did soaking change the paper, and what might slow that change?

Your turn

Write one command sentence for Little’s boat. Tell what to do.

Example: First fold the paper in half, then fold the corners down.